Individual Project Progress

AI Video Generation + 3D Gaussian Splatting

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Project Goal

Generate AI videos from text descriptions and use them to train **3D Gaussian Splatting models**

Compare results with real videos as ground truth

Timeline

Project Duration

- September - December 2025
- 6 Meetings total

Next Phase

- February 9 - June, 2026

Meetings

1. September 30
2. October 14
3. October 28
4. November 18
5. December 2
6. December 16

Early October Exploration & Setup: Key Activities (7-13)

- Researched multiple Gaussian Splatting implementations
- Successfully deployed **LichtFeld-Studio** (GUI-based, Linux)
- Learned **COLMAP** for image conversion to sparse format
- Started reading Gaussian Splatting paper
  - Detailed notes in Xournal++
- Identified GPU limitations: only 300MiB usable out of 6.1GiB

First Successful Pipeline Mid October (15-18)

Successfully generated videos in Google Colab

Models Used:

- Image: `stabilityai/stable-diffusion-2-1` (~8min)
- Video: `THUDM/CogVideoX-2b` (~25min on T4 GPU)

Critical Bug Fix:

```
# Error: CLIPTextModel.__init__() got unexpected keyword argument
# Solution:
transformers==4.49
diffusers==0.32.2
```

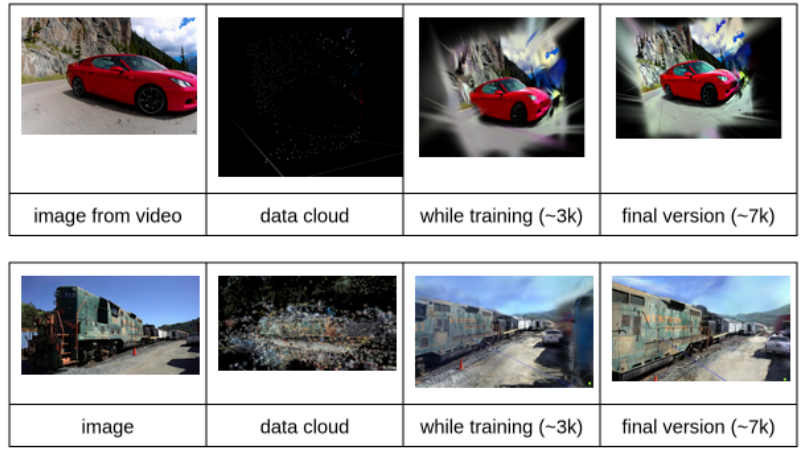
Pipeline Created: Text → Video → COLMAP → LichtFeld-Studio → 3D Model

Multi-Angle Video Experiments Results Late October (21-27)

Test	Videos	Images (COLMAP)	Images Used	Iterations	Quality
Test 1	1 (6s)	3	3	30k	Decent
Test 2	4 combined	245	61	3k-20k	Better depth
Test 3	7 combined	-	-	3k-15k	**Best result**

Test 1 (Oct 21)

- 6-second video
- 3 images extracted
- Result: "Pretty ok" for limited data
- Challenge: Training difficult with so few images



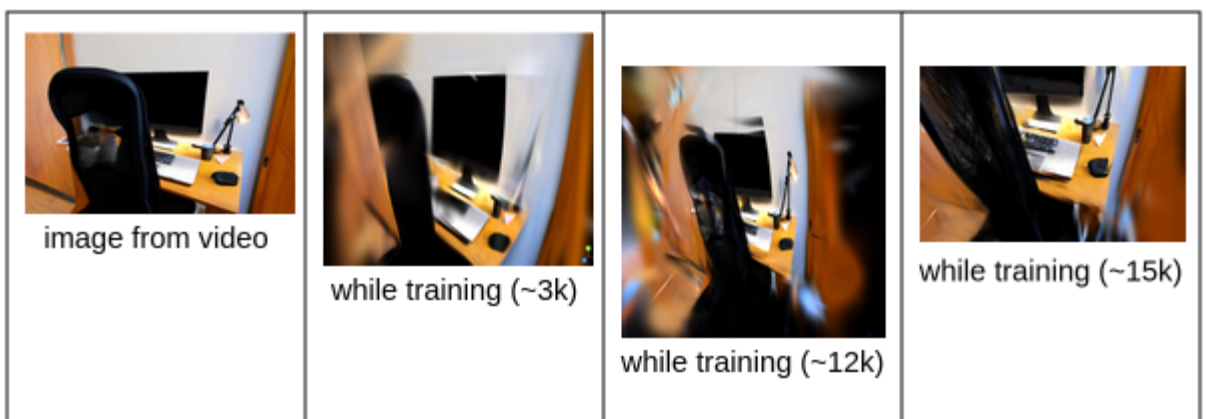
Test 2 (Oct 23)

- 4 video variations combined (~20min each)
- 245 images extracted from COLMAP
- **Only 61 images used in training**
- Result: Better depth visible
- (Completed reading Gaussian Splatting paper)

image from video	data cloud	while training (~11k)	best one

Test 3 (Oct 27)

- 7 videos from different prompts
- Combined into single video using ffmpeg
- Result: Best reconstruction so far
- Quality loss remain significant
- Prompts must be extremely precise
- Only a tiny section from the full video was used



Infrastructure Work

Docker Implementation

- Started containerization (~17GB+ data)
- Fixed LichtFeld-Studio visualization in Docker
- CUDA issues remain (ongoing)

Key Features:

- Mounted data folders for easy file transfer
- Shell scripts for automated pipeline

November (1-22) Achievements

Software Development

- Set up GitHub repository (Nov 15)
- Fixed Docker container (Nov 22)
  - Can create/clear/delete containers
  - Run commands programmatically

Academic Writing (LaTeX/Overleaf)

- Completed NeRF section (Nov 7)
- Completed 3D Gaussian Splatting section (Nov 14)

Additional Testing in November

Attempted Models:

1. `THUDM/CogVideoX-5b-I2V` (image-to-video)
   - Result: Colab timeout
2. `Facebook I2V model` (HuggingFace)
   - Status: Work in progress
3. `Tencent/hunyuanvideo-1.5`
   - Result: Needs 200GB+ storage

December Activities

- Fixed Dockerfile errors
- Created frontend/backend general structure
- **Strategic Decision:** Prioritize CLI over GUI initially
- Couldn’t work too much (exams, other projects etc)

Repositories Tested

1. `github.com/camenduru/gaussian-splatting-colab`
   - Package issues
2. **LTX-Video**
   - GPU memory limitations
3. `github.com/deepbeepmeep/Wan2GP`
   - Supports Wan 2.1/2.2, Qwen, Hunyuan, LTX, Flux
   - Too resource-intensive
4. `Wan-AI/Wan2.1-T2V-1.3B-Diffusers`
   - Needs 20GB+ RAM

More Repositories

5. `THUDM/CogVideoX-5b-I2V`
   - Timeout on Colab
6. `Tencent/hunyuanvideo-1.5`
   - Too large (200GB+)
7. `Facebook I2V model`
   - Work in progress

Critical Insights

1. **Image Count Matters**
   - Real datasets: 5,300+ images
   - AI-generated: 672 images
   - Impact: Significant quality difference
2. **Model Size Trade-off**
   - Smaller models (1-2B params) = faster
   - But: poor quality
3. **Prompt Precision**
   - T2V models need extremely detailed descriptions
   - Example: Must specify "monitor on/off"

Technical Challenges

- Package compatibility issues across T2V models
- GPU memory constraints
  - Local: 6.1GB available, only 300MB usable
- CUDA configuration in Docker
- Colab time/memory limitations

Goals

1. Test on UHasselt GPUs
   - Create videos using GPUs to have something to work with
   - Generate using different techniques
2. Compare the different techniques
3. Finish paper
   - Can’t be finished without having some results
4. Prepare end-of-semester presentation

Frontend using gradio - 1

🎥 LichtFeld-Studio Video Processor

Process videos through the LichtFeld-Studio pipeline to generate 3D reconstructions.

Pipeline Steps:

1. **COLMAP Pipeline**: Extracts frames and runs COLMAP reconstruction
2. **LichtFeld-Studio**: Opens GUI for training and model generation
3. **Output**: Downloads all generated files (including PLY files)

📋 Input Configuration

Video File

output.mp4 2.7 MB

Project Directory Name

reconstruction_project

Frames Per Second (FPS)

30

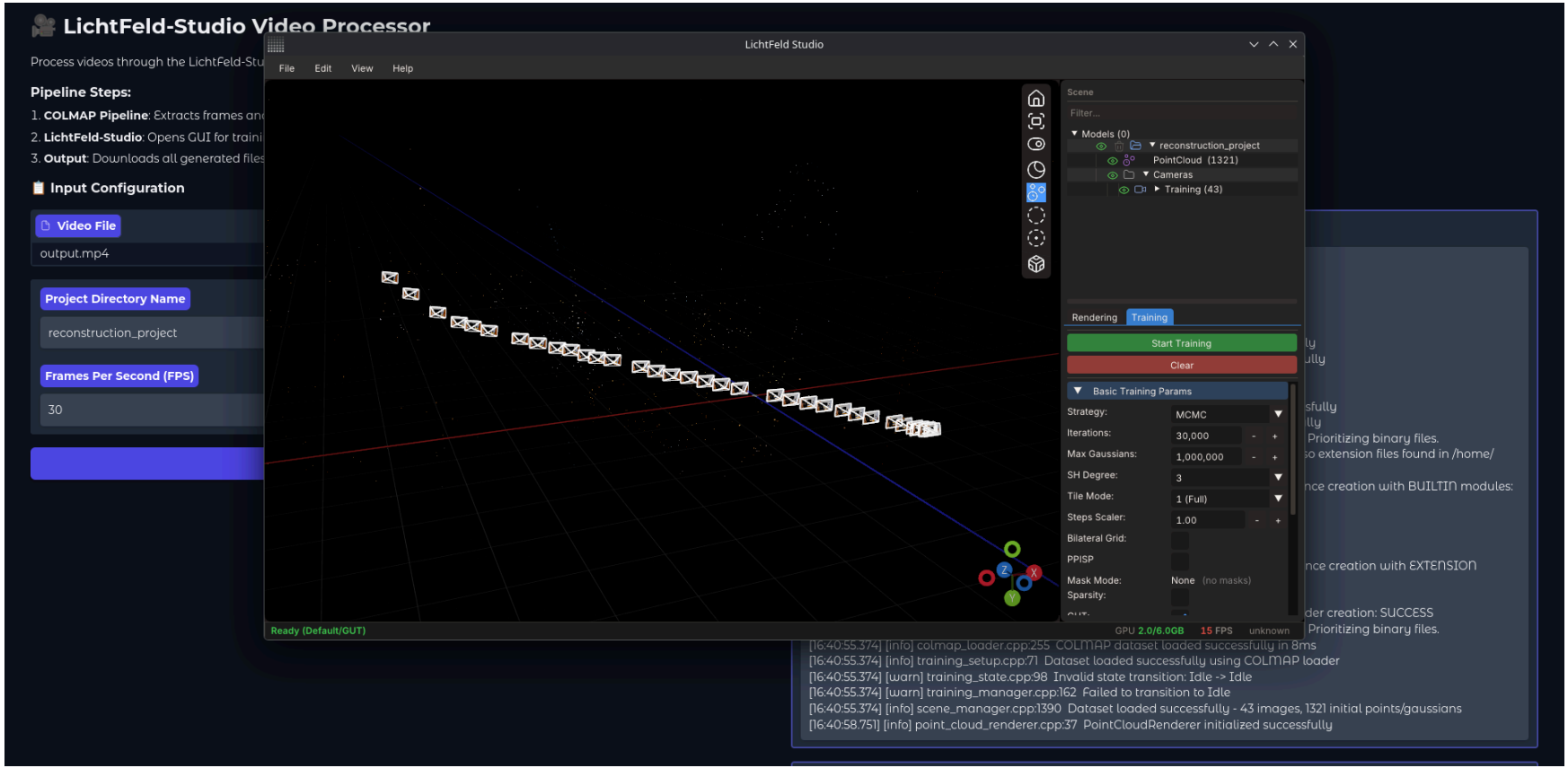
🚀 Start Processing

📊 Processing Status

Status (Important messages only)

```
------------------------------------------------------------
🔄 STEP 2: Running LichtFeld-Studio
------------------------------------------------------------
⚠️ GUI window will open - train and close when done

[16:40:55.357] [info] bbox_renderer.cpp:105  Bounding box renderer initialized successfully
[16:40:55.358] [info] axes_renderer.cpp:130  Coordinate axes renderer initialized successfully
[16:40:55.359] [info] text_renderer.cpp:125  Successfully loaded 3 characters from font
[16:40:55.359] [info] viewport_gizmo.cpp:62  ViewportGizmo initialized successfully
[16:40:55.362] [info] rendering_engine_impl.cpp:102  Rendering engine initialized successfully
[16:40:55.362] [info] rendering_manager.cpp:328  Rendering engine initialized successfully
[16:40:55.362] [warn] colmap_loader.cpp:73  Found both binary and text COLMAP files. Prioritizing binary files.
[16:40:55.363] [warn] nvcodec_image_loader.cpp:172  [nvImageCodec Diagnostics] No .so extension files found in /home/
projects/LichtFeld-Studio/build/extensions
[16:40:55.363] [info] nvcodec_image_loader.cpp:259  [nvImageCodec Diagnostics] Instance creation with BUILTIN modules:
SUCCESS
[16:40:55.364] [warn] training_state.cpp:98  Invalid state transition: Idle -> Idle
[16:40:55.364] [warn] training_manager.cpp:162  Failed to transition to Idle
[16:40:55.364] [warn] memory_arena.cu:360  Emergency cleanup started
[16:40:55.364] [info] nvcodec_image_loader.cpp:274  [nvImageCodec Diagnostics] Instance creation with EXTENSION
modules (/home/projects/LichtFeld-Studio/build/extensions): SUCCESS
[16:40:55.364] [warn] memory_arena.cu:388  Emergency cleanup completed
[16:40:55.366] [info] nvcodec_image_loader.cpp:309  [nvImageCodec Diagnostics] Decoder creation: SUCCESS
[16:40:55.366] [warn] colmap_loader.cpp:73  Found both binary and text COLMAP files. Prioritizing binary files.
[16:40:55.374] [info] colmap_loader.cpp:255  COLMAP dataset loaded successfully in 8ms
[16:40:55.374] [info] training_setup.cpp:71  Dataset loaded successfully using COLMAP loader
[16:40:55.374] [warn] training_state.cpp:98  Invalid state transition: Idle -> Idle
[16:40:55.374] [warn] training_manager.cpp:162  Failed to transition to Idle
[16:40:55.374] [info] scene_manager.cpp:1390  Dataset loaded successfully - 43 images, 1321 initial points/gaussians
[16:40:58.751] [info] point_cloud_renderer.cpp:37  PointCloudRenderer initialized successfully
```

Frontend using gradio - 2



Frontend using gradio - 3

Pipeline Steps:

1. **COLMAP Pipeline**: Extracts frames and runs COLMAP reconstruction
2. **LichtFeld-Studio**: Opens GUI for training and model generation
3. **Output**: Downloads all generated files (including PLY files)

📋 **Input Configuration**

Video File

output.mp4 2.7 MB

Project Directory Name

reconstruction_project

Frames Per Second (FPS)

30

🚀 **Start Processing**

📊 **Processing Status**

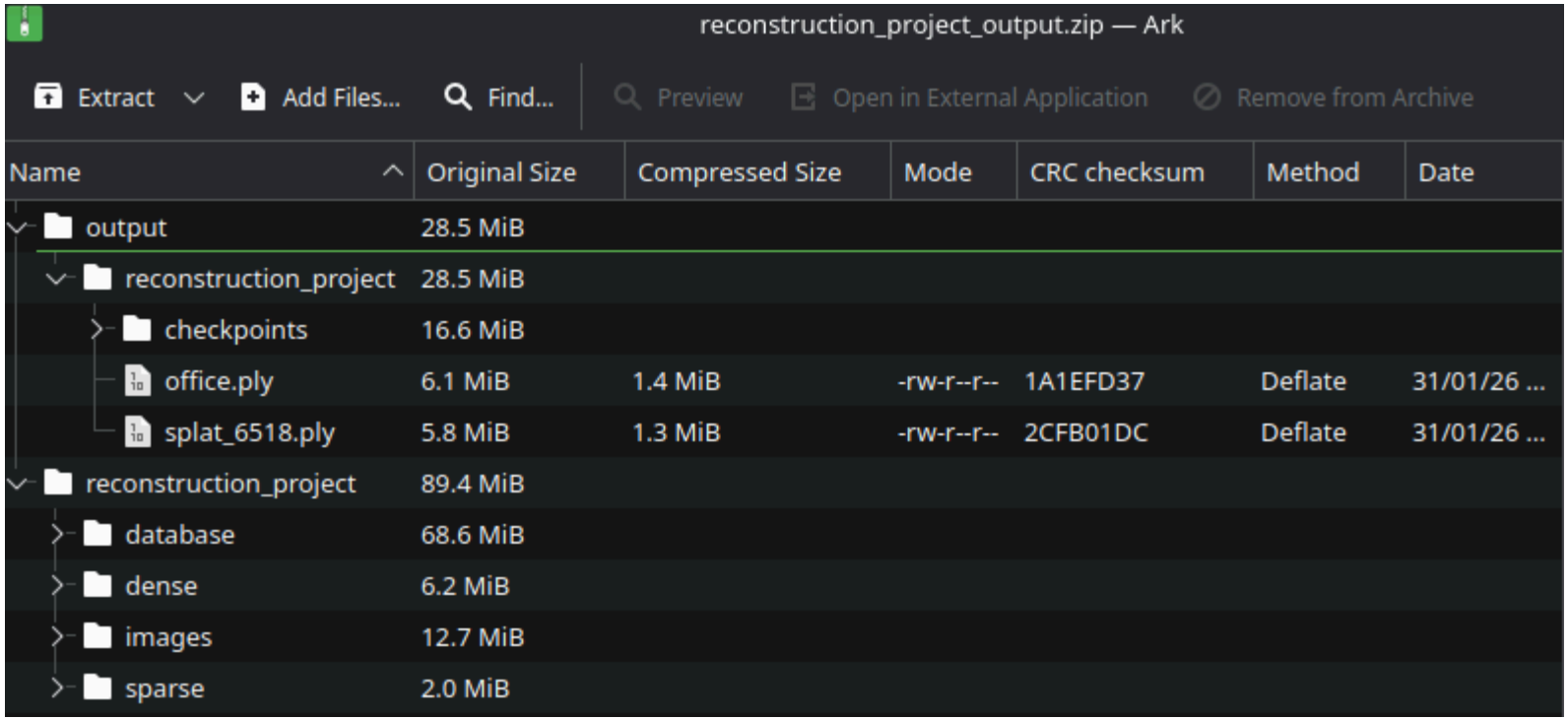
Status (Important messages only)

```
[16:42:29.513] [warn] nvcodec_image_loader.cpp:172  [nvImageCodec Diagnostics] No .so extension files found in /home/
projects/LichtFeld-Studio/build/extensions
[16:42:29.514] [info] nvcodec_image_loader.cpp:259  [nvImageCodec Diagnostics] Instance creation with BUILTIN modules:
SUCCESS
[16:42:29.515] [info] nvcodec_image_loader.cpp:274  [nvImageCodec Diagnostics] Instance creation with EXTENSION
modules (/home/projects/LichtFeld-Studio/build/extensions): SUCCESS
[16:42:29.517] [info] nvcodec_image_loader.cpp:309  [nvImageCodec Diagnostics] Decoder creation: SUCCESS
⏳ Training... (145 operations)
⏳ Training... (149 operations)
⏳ Training... (154 operations)
⏳ Training... (164 operations)
[16:45:30.735] [info] gui_manager.cpp:2998  Export completed: /home/projects/LichtFeld-Studio/output/
reconstruction_project/office.ply
[16:45:35.966] [info] trainer.cpp:1703  Training completed successfully
[16:45:35.969] [info] training_manager.cpp:521  Training completed successfully
[16:45:36.028] [warn] memory_arena.cu:360  Emergency cleanup started
[16:45:36.028] [warn] memory_arena.cu:388  Emergency cleanup completed
[16:45:36.028] [info] image_io.cpp:570  [BatchImageSaver] Shutdown complete

⚠️ GUI closed with segfault (exit -11), but training completed successfully
✅ Treating as successful completion

✅ LichtFeld-Studio completed!
============================================================

🔄 STEP 3: Creating Download Archive
------------------------------------------------------------
 Archived 100 files...
 Archived 200 files...
 Archived 300 files...
```

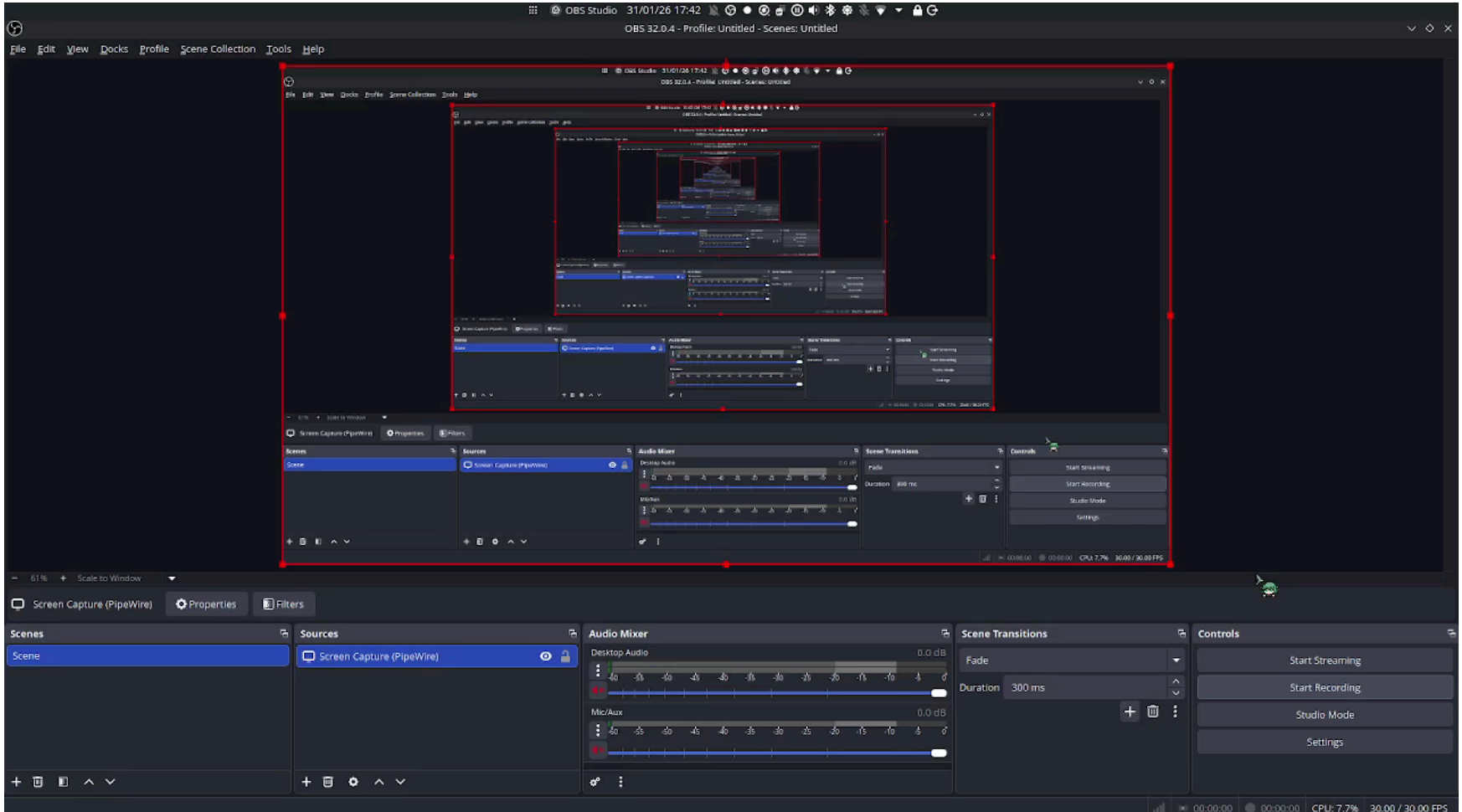
Download Results

reconstruction_project_output.zip 59.2 MB

Frontend using gradio - 4



Name	Original Size	Compressed Size	Mode	CRC checksum	Method	Date
output	28.5 MiB					
reconstruction_project	28.5 MiB					
checkpoints	16.6 MiB					
office.ply	6.1 MiB	1.4 MiB	-rw-r--r--	1A1EFD37	Deflate	31/01/26 ...
splat_6518.ply	5.8 MiB	1.3 MiB	-rw-r--r--	2CFB01DC	Deflate	31/01/26 ...
reconstruction_project	89.4 MiB					
database	68.6 MiB					
dense	6.2 MiB					
images	12.7 MiB					
sparse	2.0 MiB					


OBS Studio 31/01/26 17:42
OBS 32.0.4 - Profile: Untitled - Scenes: Untitled
File Edit View Docks Profile Scene Collection Tools Help
Screen Capture (PipeWire)
Properties
Filters
Scenes
Scene
Sources
Screen Capture (PipeWire)
Audio Mixer
Desktop Audio 0.0 dB
Mic/Aux 0.0 dB
Scene Transitions
Fade
Duration 300 ms
Controls
Start Streaming
Start Recording
Studio Mode
Settings
00:00:00
00:00:00
CPU: 7.7%
30.00 / 30.00 FPS